

Homework 06 Solutions

Question 1: Compare KNN and logistic regression

Solution

▼ Show the solution code

```
library(FNN)
library(pROC)

data_dir <- if (file.exists("data/breast-cancer.csv")) {
  "data"
} else {
  "practice/weeks/week-06/data"
}
data <- read.csv(file.path(data_dir, "breast-cancer.csv"))
set.seed(432)
train_id <- sort(sample(seq_len(nrow(data)), floor(0.7 * nrow(data))))
train <- data[train_id, ]
test <- data[-train_id, ]
predictors <- c("mean_radius", "mean_texture", "mean_smoothness")

x_bar <- colMeans(train[, predictors])
s <- sapply(train[, predictors], sd)
X_train <- scale(train[, predictors], center = x_bar, scale = s)
X_test <- scale(test[, predictors], center = x_bar, scale = s)
y_train <- train$y
y_test <- test$y
```

▼ Show the solution code

```
fit <- glm(y ~ ., data = data.frame(y = y_train, X_train),
          family = binomial())
k_values <- c(5, 15, 25, 35, 45)
model_names <- c("Logistic regression", paste("KNN, k =", k_values))
p_test <- matrix(NA_real_, nrow(X_test), length(model_names))
p_test[, 1] <- predict(fit, newdata = data.frame(X_test), type = "response")
for (m in seq_along(k_values)) {
  p_test[, m + 1] <- knn.reg(train = X_train, test = X_test,
                             y = y_train, k = k_values[m],
                             algorithm = "brute")$pred
}

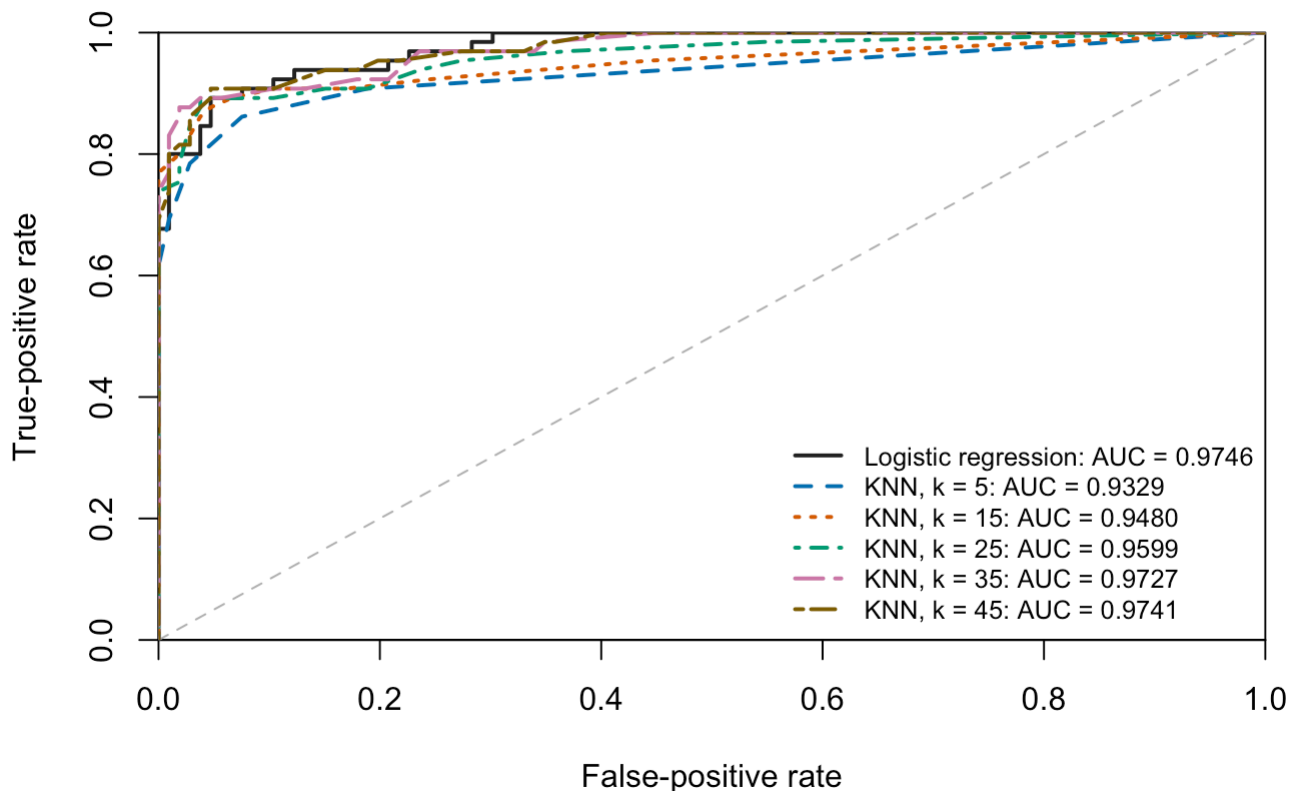
roc_test <- vector("list", length(model_names))
auc_test <- numeric(length(model_names))
for (m in seq_along(model_names)) {
  roc_test[[m]] <- roc(y_test, p_test[, m], levels = c(0, 1),
                      direction = "<", quiet = TRUE)
  auc_test[m] <- as.numeric(auc(roc_test[[m]]))
}
```

```
}
data.frame(Model = model_names, Test_AUC = round(auc_test, 4))
```

	Model	Test_AUC
1	Logistic regression	0.9746
2	KNN, k = 5	0.9329
3	KNN, k = 15	0.9480
4	KNN, k = 25	0.9599
5	KNN, k = 35	0.9727
6	KNN, k = 45	0.9741

▼ Show the solution code

```
#| fig-width: 6.5
#| fig-height: 6
#| fig-cap: "Test ROC curves for logistic regression and five KNN fits on the same 171
#| fig-alt: "Six test ROC curves, with their AUC values in the legend, compared on com
colors <- c("#222222", "#0072B2", "#D55E00", "#009E73", "#CC79A7", "#806000")
line_types <- c(1, 2, 3, 4, 5, 6)
plot(c(0, 1), c(0, 1), type = "n", xaxs = "i", yaxs = "i",
     xlab = "False-positive rate", ylab = "True-positive rate")
abline(0, 1, col = "gray70", lty = 2)
for (m in seq_along(model_names)) {
  lines(1 - roc_test[[m]]$specificities, roc_test[[m]]$sensitivities,
       col = colors[m], lty = line_types[m], lwd = 2)
}
legend("bottomright", sprintf("%s: AUC = %.4f", model_names, auc_test),
     col = colors, lty = line_types, lwd = 2, cex = 0.8, bty = "n")
```



Based on these results, I prefer logistic regression because it has the largest AUC test score with 0.9746, indicating better predictive performance.

Question 2: Training and test ROC curves

Solution

▼ Show the solution code

```
k_values <- c(1, 3, 5, 10, 20)
p_train <- matrix(NA_real_, nrow(X_train), length(k_values))
p_test <- matrix(NA_real_, nrow(X_test), length(k_values))
roc_train <- roc_test <- vector("list", length(k_values))
auc_train <- auc_test <- numeric(length(k_values))
for (m in seq_along(k_values)) {
  k <- k_values[m]
  # test = X_train includes each training observation itself.
  p_train[, m] <- knn.reg(train = X_train, test = X_train,
                          y = y_train, k = k, algorithm = "brute")$pred
  p_test[, m] <- knn.reg(train = X_train, test = X_test,
                         y = y_train, k = k, algorithm = "brute")$pred
  roc_train[[m]] <- roc(y_train, p_train[, m], levels = c(0, 1),
                       direction = "<", quiet = TRUE)
  roc_test[[m]] <- roc(y_test, p_test[, m], levels = c(0, 1),
                      direction = "<", quiet = TRUE)
  auc_train[m] <- as.numeric(auc(roc_train[[m]]))
}
```

```

    auc_test[m] <- as.numeric(auc(roc_test[[m]]))
  }
  data.frame(k = k_values, Training_AUC = round(auc_train, 4),
            Test_AUC = round(auc_test, 4))

```

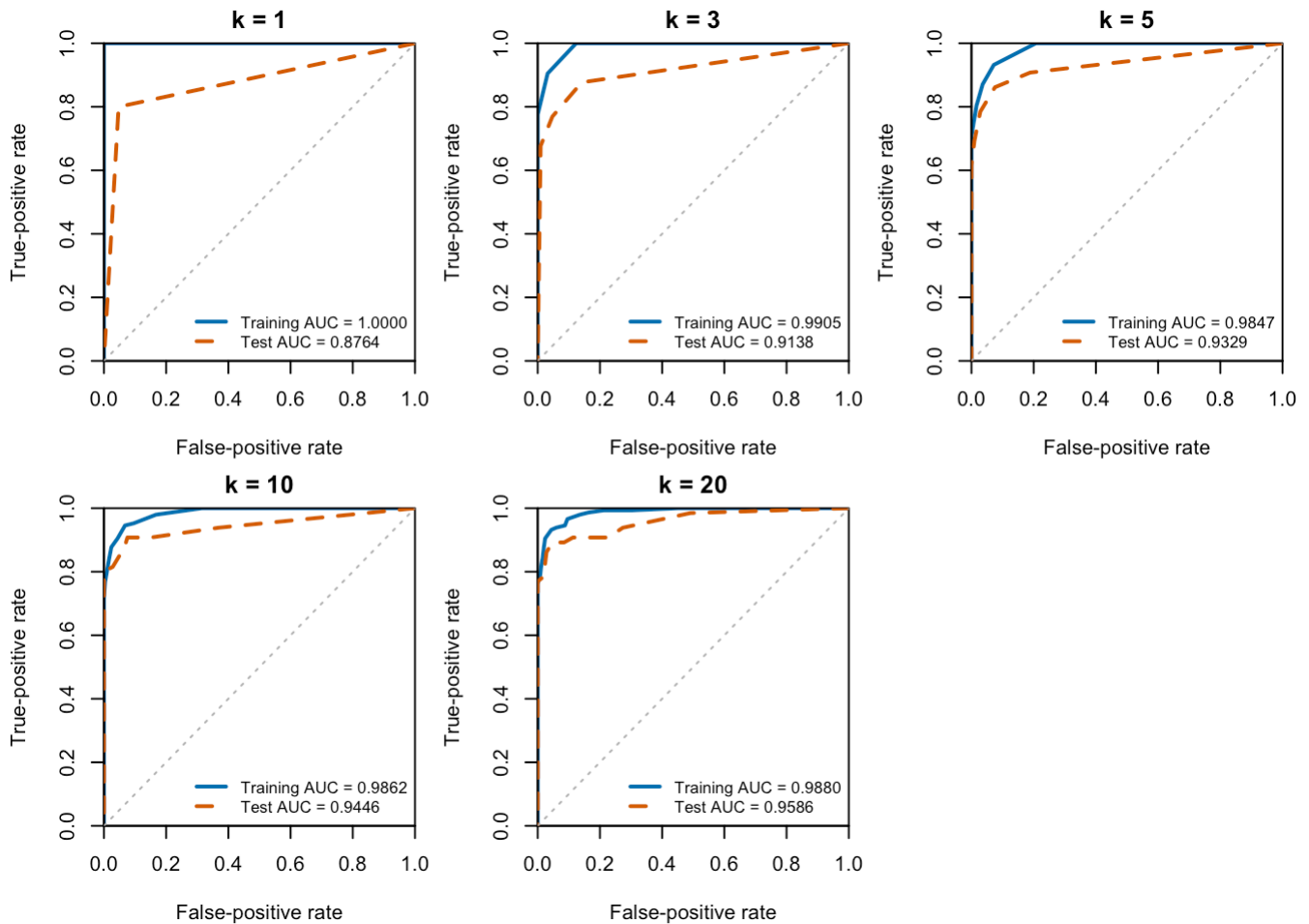
	k	Training_AUC	Test_AUC
1	1	1.0000	0.8764
2	3	0.9905	0.9138
3	5	0.9847	0.9329
4	10	0.9862	0.9446
5	20	0.9880	0.9586

▼ Show the solution code

```

#| fig-width: 10
#| fig-height: 7
#| fig-cap: "Training and test ROC curves for each KNN neighborhood size. Each trainin
#| fig-alt: "Five panels compare training and test ROC curves for k equal to 1, 3, 5,
old_par <- par(mfrow = c(2, 3), mar = c(4, 4, 2, 1))
for (m in seq_along(k_values)) {
  plot(1 - roc_train[[m]]$specificities, roc_train[[m]]$sensitivities,
       type = "l", col = "#0072B2", lwd = 2,
       xlim = c(0, 1), ylim = c(0, 1), xaxs = "i", yaxs = "i",
       main = paste("k =", k_values[m]),
       xlab = "False-positive rate", ylab = "True-positive rate")
  lines(1 - roc_test[[m]]$specificities, roc_test[[m]]$sensitivities,
        col = "#D55E00", lty = 2, lwd = 2)
  abline(0, 1, col = "gray70", lty = 3)
  legend("bottomright", c(sprintf("Training AUC = %.4f", auc_train[m]),
                              sprintf("Test AUC = %.4f", auc_test[m])),
        col = c("#0072B2", "#D55E00"), lty = c(1, 2), lwd = 2,
        cex = 0.8, bty = "n")
}
plot.new()

```



▼ Show the solution code

```
par(old_par)
```

When $K = 1$, the training AUC score is 1, which shows a perfect fit in the training data. But their testing AUC score is only 0.8764, which indicates a lower predictive performance. However, when K increases, the testing AUC started to grows as well, and ends at 0.9586 when $K = 20$. But when K increases, the training AUC decreases to 0.9847 ($K=5$), and the increases until 0.9880 when $K = 20$.

Question 3: Peer evaluation of Homework 5

Feedback1: <https://github.com/zexuanj/stat432-fall2026/issues/2> Feedback2: <https://github.com/mirgaon314/stat432-fall2026/issues/4>